

Chapter 169

Resolve FX Warp

The filters in this category let you warp the image in different ways, creating procedural or custom distortion effects, some of which can be automatically animated. The Warper plugin is a point-based free-form Warp tool.

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Dent

A warp effect that creates different types of circular bowing and folding effects.

- **DentType:** This lets you choose from six different types of “dent” warping effects. The options are Type 1, Type 2, Type 3, Sine, Cosine, and Black Hole.
- **X and Y:** Position sliders let you offset the center of the warp.
- **Size:** This lets you adjust the diameter of the warp.
- **Strength:** This lets you adjust the extent and direction of the warp, lowering this below zero “pulls” the image into the center of the effect, while raising this above zero “pushes” the image outward to buckle according to the DentType you’ve chosen.
- **Edge Behavior:** This eliminates any blanking caused by negative distortion by filling black areas in one of several user-defined ways; options include Black (do nothing and leave the blanking), Reflect, Wrap-Around, and Replicate.

Lens Distortion (Studio Version Only)

Lets you add convex or concave lens distortion to an image, to make it warp by bulging outward or bulging inward, similar to a poorly made lens. This filter could also be used in a corrective manner, to compensate for images exhibiting barrel distortion. Additionally, the amount of distortion applied to each color channel can be varied, in order to create the effect of chromatic aberration using the curvature of the distortion you’re adding for a more accurate effect.

- **Red, Green, and Blue Distortion:** These sliders are ordinarily locked together by the Gang checkbox. When adjusted as ganged, moving these sliders to the left (negative values) increases “fisheye” distortion, while moving these sliders to the right (positive values) creates an inverted “fisheye” effect that bows the image inward rather than outward. Turning Gang off lets you set the Red, Green, and Blue Distortion sliders to different values, introducing more accurately simulated chromatic aberration to the image.
- **Fine Adjustment:** This checkbox results in the distortion sliders operating within a much smaller range, enabling more precise adjustments.
- **X and Y Position:** These sliders let you change the center of the distortion effect.
- **Edge Behavior:** The options available in this pop-up eliminate blanking caused by negative distortion by filling black areas in one of several user-defined ways; options include Black (do nothing and leave the blanking), Reflect, Wrap-Around, and Replicate.

TIP: To resize a clip with an extreme distortion effect that introduces blanking around the edges of the frame, you’ll need to use the Zoom control found in the Node Sizing mode of the Sizing palette. Edit and Input sizing will only zoom the image within the border created by blanking.

Ripples

A warp effect that creates different types of ripples.

Main Controls

These controls govern the type of ripple and the quality of optional ripple shine that you want.

- **Ripple Shape:** Lets you choose from Circular, Square, Horizontal, Vertical, Exponential, Star, and Radial ripples.
- **Wave Shape:** Lets you choose from Sinusoidal, Triangular, Fresnel In, Fresnel Out, and Natural waves.

A set of Shine parameters lets you overlay a simulated shine on top of these ripples.

- **Shine:** A checkbox lets you turn this shine on and off.
- **Shine Direction:** Lets you add a light shine to the peaks of the ripples. Works best at low shine strength.
- **Vertical Shine Height:** Lets you adjust the thickness of the highlights running along the ripples.
- **Shine Size:** Lets you adjust the length of the highlights running along the ripples.
- **Shine Strength:** Lets you adjust the overall appearance of the shine.
- **Animate:** This checkbox, if checked, reveals a Speed slider that lets you set how quickly the ripple effect auto-animates the phase of the currently configured ripple effect, without the need to use keyframes.

Ripple Controls

A separate set of Ripple parameters lets you enable up to five overlapping ripples.

- **Enable Ripple:** A keyframable checkbox that will turn a particular ripple on or off.
- **X and Y Position:** Lets you change the center position of each set of ripples.
- **Amplitude:** Lets you adjust the “height” of each ripple caused by this effect.
- **Frequency:** Lets you choose how many ripples are being created.
- **Decay:** Lets you adjust the falloff of the ripple effect, where ripples gradually diminish as they reach the point of decay.
- **Phase:** Lets you adjust the phase of the ripple effect relative to the center origination point. If you want to animate a ripple simply, you can also choose to keyframe the Phase parameter to create the same effect created by the Animate checkbox above.

Vortex

Vortex begins as an S shaped warp effect, but you can adjust the parameters to create many types of warping effects.

- **X and Y position:** Two sliders let you offset the center of the warp.
- **Size:** Adjusts the diameter of the warp.
- **Angle:** Adjusts the direction and intensity of the warp effect; setting Angle below 0 twists the image to the left, while raising Angle over 0 twists the image to the right.
- **Power:** Rotates the area that’s affected to create a harder border between what is warped and what isn’t.
- **Swirl:** Intensifies the effect, twisting the image into an even tighter swirling effect.
- **Edge Behavior:** The options available in this pop-up eliminate blanking caused by negative distortion by filling black areas in one of several user-defined ways; options include Black (do nothing and leave the blanking), Reflect, Wrap-Around, and Replicate.

Warper (Studio Version Only)

The Warper is a free-form image warper that uses points or curves to push and stretch features in an image as if they were on a sheet of rubber. While the Warper has a lot of options and settings, getting started is really easy.

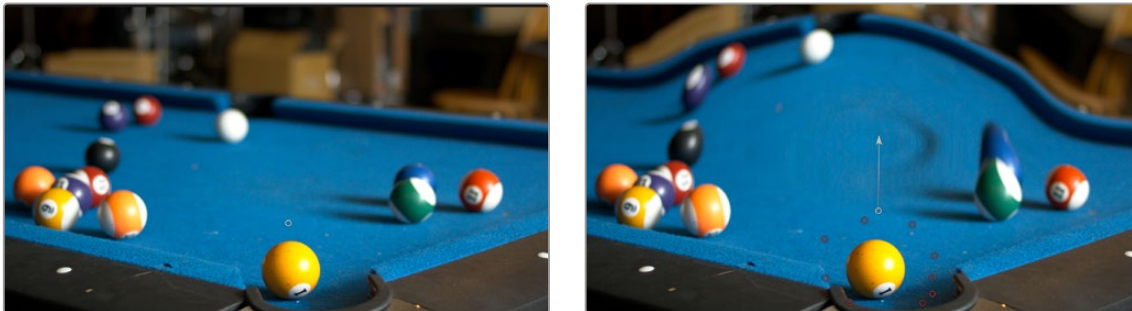
Warp Mode Points

To warp a feature of the image, just select Points as the Warp mode, click anywhere in the image to add a warp point, and drag to push that part of the image in the direction you want it to go. Warp points are gray. By default, warp points influence the entire image, which is pinned down at the corners. This system makes it simple to warp large regions of the image, such as when you want to warp an edge of the image to get rid of a boom dip.



(Left) The original image, (Right) The image warped with a single control point

To “pin down” parts of the image you don’t want to warp, Shift-click to place limiter points (which are red). By combining warp points with limiter points, you can quickly create extremely specific warp effects to squish and stretch features in an image in any way you want.



(Left) The original image, (Right) The image warped with control (gray) and limiter (red) points

As you work, you can Option-click any unwanted warp or limiter point to delete it. To start from scratch, you can click the Reset Warp control in the Inspector. Warp and limiter points can be keyframed via the Positions and Warp Strength parameters.

Warp Mode Curves

The Curves mode of the Warper lets you generate the warp around a curve or shape, instead of simple points on a grid. This lets you get more specific with your selection and make a more organic-looking warp on the image. The Curves mode adds more drawing tools and keyboard shortcuts to the Viewer.

To draw a curve, first select Warp Mode Curves, then click in the Viewer where you want the curve to start, then drag the line to where you want the next point in the curve and click again. Then continue dragging lines and clicking points until you have the rough line or shape you need. Curves can be used open or closed into shapes.

Add, Adjust, and Warp modes

When creating a curve, initially you will be in Add mode; this allows you to create your initial curve or shape, however once you've closed the shape or pressed Esc to stop drawing, by default the Warper will automatically shift you to Adjust mode, which allows you to manipulate your curve without adding more points to it. You then switch to Warp mode to move the initial curve points to another location, which actually performs the warp. You can have as many curves or shapes as you can draw, and if one curve intersects another, a point will automatically be placed at the intersection.

You can manually toggle back and forth between Add, Adjust, and Warp modes as needed.

Curve Drawing Commands

Most of the curve editing is done intuitively by clicking and dragging in the Viewer, while there are also some additional keyboard commands.

- **Esc:** Stops drawing.
- **Enter:** Closes the curve connecting the last point to the first point, or simply click on the first point again to close the shape.
- **Click:** Anywhere in the image to start a new curve.
- **Click:** Anywhere on an existing curve to add another point.
- **Click and Drag:** To set a point and drag out its spline handles to adjust its curvature.
- **Shift-Click:** Anywhere in the image to start a limiter curve. A limiter curve lets you define a shape or area where no warping will take place. A limiter curve will be in red.
- **Option-Click:** Creates a new line in Add mode, regardless of if you are currently in Adjust or Warp modes.
- **Delete:** Removes a selected point or points.



Drawing curves in the Warper

In the above example, the limiter (red) shape around the 1 ball prevents it from being warped. The straight line on the left is warping the table downward and to the left along its edge, as seen by the origin reference (dotted lines) showing where the warp began and the solid line showing where the warp is. The curve over the group of balls to the right is showing the spline handles of the currently selected point.

Editing Splines

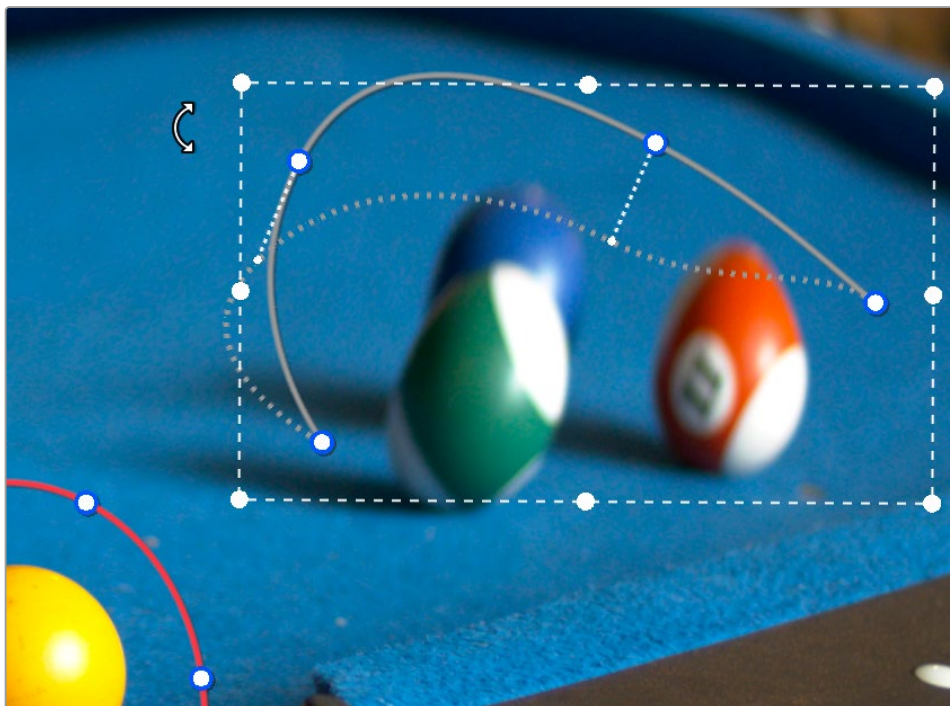
By default, your “curves” will look more like sharp lines rather than graceful flowing curves. You can manipulate the smoothness of the curve by adjusting a point’s spline handles.

A point’s spline handle is seen by clicking directly on the point. If the handles don’t appear, hold down the Command key and drag the handles out of the point. By default spline handles are paired, as you move one, the other moves in the opposite direction. You can also manipulate one spline handle at a time by holding the Command key and clicking on the single spline handle you want to manipulate.

Selection

To select curves and points for modification, you can click directly on them in viewer. Click on the shape again to enclose them in a bounding box. You can select either the whole shape, a point, or drag to draw a selection box around certain points. Holding shift and drag to creates a freeform selection lasso instead of a box.

Once the box is in place, you can scale the points by dragging the bounding box from it’s edges. You can rotate the selection by hovering the mouse near the edge of the box until the rotate tool appears, then click and drag to rotate. Holding Option and dragging scales the shape around the geometric center of the box.



Drawing a bounding box around a curve, lets you scale and rotate it.

Warper Settings

Warp Mode: Choose which control mode to apply the warp, either using Curves or Points.

Curves

- **Add:** With this mode selected, you can add points to create a curve or shape to warp around.
- **Adjust:** With this mode selected, you can adjust the curve or shape that you've created isolated from the warp.
- **Warp:** With this mode selected, the image will warp around the curves or shapes that you've created and update in real time as you move them.
- **Render a Reference Grid:** Check this box to add a reference grid over the image, so you can see just how much the warp is affecting each area of the frame.
- **Origin Reference:** Adjusting this slider will adjust the visible intensity of the original curve and point locations. Slide to the right to increase the intensity, slide to the left to make them transparent.
- **Position Keyframing:** Keyframes the positions of the curve points.
- **Warp Keyframing:** Keyframes the warps of the curve points.
- **Warp Strength:** Adjust this slider to magnify or reduce the distortion effect of the warp.

Selected Points

- **Symmetric:** Makes the selected point or all points on selected curve symmetric to the selected control point.
- **Corner:** Makes the selected point or all points on selected curve hard corners.
- **Rounded:** Makes the selected point or all points on selected curve rounded corners with spline controls.
- **Reset Handle Adjustments (Warp mode):** Click this button to reset the spline curve adjustments of a point back to neutral.
- **Reset Warp (Warp mode):** Click this button to reset the warp back to its original neutral state.

Selected Curve

- **Show All Handles:** This checkbox toggles on or off all the spline handles for all the points on a curve.
- **Open/Close:** Click this button to toggle closing a curve (connecting the last point to the first point) or opening a curve (removing the connection).
- **Warper/Limiter:** Click this button to toggle a curve between a warper or limiter curve. A limiter curve prevents warping in its defined area.
- **Delete:** Deletes an entire curve.

Points

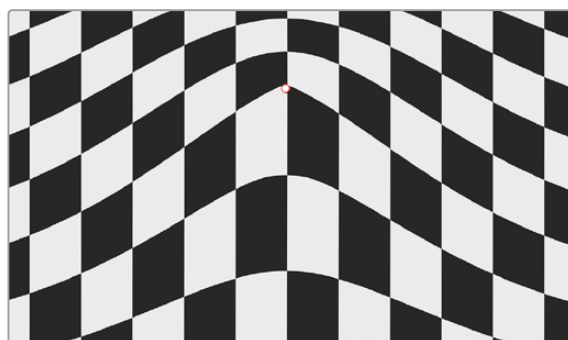
- **Render a Reference Grid:** Check this box to add a reference grid over the image, so you can see just how much the warp is affecting each area of the frame.
- **Positions:** Keyframes the positions of the points in the Viewer.
- **Show Origin Locations:** Shows the original location of each point in the Warper.
- **Clear All Points:** Deletes all added points and returns the Warper to its neutral state.
- **Reset Warp:** Resets the Warper to its neutral state but retains all added points at their origin position.
- **Warp Strength:** Adjust this slider to magnify or reduce the distortion effect of the warp.

On-Screen Controls

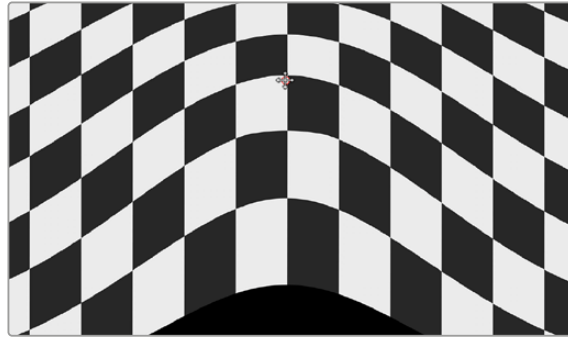
- **Show:** Shows the on-screen controls in the Viewer.
- **Auto Hide:** Shows the on-screen controls in the Viewer but hides them only as they are adjusted, letting you see the final image while making adjustments.
- **Hide:** Hides the on-screen controls in the Viewer.
- **Warp Boundaries (Points Only):** This pop-up menu has three choices for letting you see which areas of the image will be locked off by limiter points you place in the image.
 - None:** Hides this information and is the default setting.
 - As Border:** Shows a red line at the boundary where warping becomes locked off.
 - As Mask:** Dims the area that's protected from warping by limiter points.
- **Scale Controls:** This slider lets you shrink or grow the control points or warp vector handles to the most convenient size for the task at hand.

Advanced Options

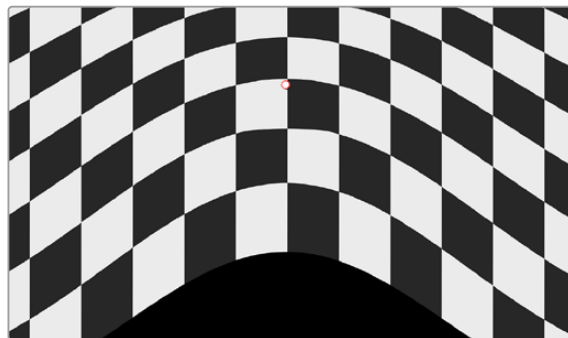
- **Limits:** A pop-up menu lets you choose how to treat the edges of the image when you're warping it.
 - Pin at Corners:** Locks the four corners of the image in place, but allows the sides to bulge in or out as you warp and allows blanking to creep in.
 - Around the Edges:** Locks the full width and height of all four edges in place as you warp, preventing blanking.
 - Distant:** Locks parts of the image that are n pixels away, but the default setting is very permissive.
 - Manual (Free-Floating):** Allows the entire image to be warped without locking any part of it, allowing for extreme warps all along an entire edge of the image, but requiring you to place at least one limiter point prior to adding a warp point.
- **Elasticity:** A pop-up menu lets you choose one of three methods of interpolating how the image bends around a Warp point: Fabric, Rubber (the default), and Jelly. Fabric produces the “most pointy” warps, while Jelly produces the most gently curved warps. Rubber strikes a balance between the two extremes..



Fabric



Rubber



Jelly

- **Edge Behavior:** A pop-up menu eliminates any blanking caused by warping that affects the edge of the image by filling black areas in one of several user-defined ways; options include Black (do nothing and leave the blanking), Reflect, Wrap-Around, and Replicate.
- **Warp Detail:** A pop-up menu lets you choose from among three methods of calculating the warping effect you're creating: Faster, Default, and Better. Each choice is an obvious trade-off between image smoothness and performance.
- **Effect Follows:** Allows you to choose whether the points stick to the scaled content (Input and Edit Sizing) or the original frame in the timeline (Timeline Frame).
- **Auto Advance to Adjust Mode:** When this box is checked, after adding a point or curve, automatically switch to Adjust mode to edit it.

Waviness

- **Waviness Type:** Lets you choose Vertical or Horizontal waviness.
- **Scale:** Lets you choose the size of the waves.
- **Strength:** Lets you adjust the amplitude.
- **Phase:** Lets you "move" the wave and is a good parameter to keyframe if you're animating this effect manually.
- **Animate:** A checkbox that reveals a Speed slider that lets you set how quickly the Waviness effect auto-animates without the need to use keyframes.



Fairlight

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